Fair Carbon-Aware Multi-Agent Reinforcement Learning for Building Energy Management

Application Project

Project Report

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1 Introduction

1.1 Background

The transition towards low-carbon energy systems requires effective management of both electricity supply and demand. Buildings can contribute to this transition through distributed energy resources such as photovoltaic generation, batteries, thermal storage, and controllable heating and cooling systems. These resources provide flexibility by allowing electricity consumption to be shifted across time or adjusted in response to changing grid conditions. However, their operation must account for building requirements and occupant comfort. Coordinating flexible resources across multiple buildings therefore creates a sequential decision-making problem involving energy consumption, environmental objectives, and service delivery [1, 2].

Reinforcement learning (RL) provides an approach to this problem by learning control policies through interaction with an environment. An RL controller observes the current building state, selects actions, and receives rewards that reflect its control objectives. Its decisions can account for delayed consequences, such as how charging a battery or changing cooling power affects future electricity demand and indoor temperature. CityLearn provides a common simulation environment for implementing and comparing such controllers across individual buildings and building communities [1].

Among the available methods, Soft Actor-Critic (SAC) supports continuous control through an off-policy actor-critic architecture and an entropy-augmented learning objective [3]. Independent SAC assigns a separate policy to each building. Multi-Agent Reinforcement Learning with Iterative Sequential Action Selection (MARLISA) extends this approach by introducing demand estimation, information sharing, and sequential action selection to support coordination across buildings [4]. These methods provide a foundation for examining how local control decisions affect district-level outcomes.

1.2 Problem Statement

An effective building controller must balance environmental performance with the services provided to occupants. Minimizing electricity consumption or operational carbon emissions alone may encourage actions that reduce cooling provision and increase discomfort. Conversely, maintaining comfort without considering energy use may increase electricity demand and emissions. The control problem consequently involves competing objectives whose relative importance depends on the reward function.

A further concern is how these outcomes are distributed across buildings. Improvements in district-level performance do not necessarily imply that every building benefits. Differences in occupancy, thermal characteristics, electricity demand, and available equipment can produce unequal comfort outcomes under a common control strategy. Average discomfort may therefore conceal a building that experiences substantially worse service than the others.

Fairness also requires more than minimizing differences between buildings. A controller can produce a small discomfort gap when all buildings experience similarly poor comfort. Evaluating fairness consequently requires examining both the quality of service and its

distribution. This project addresses that problem by considering average discomfort, worst-building discomfort, and the gap between the highest and lowest building discomfort proportions alongside carbon emissions and energy-related indicators.

1.3 Project Objective

The overall objective of this project is to investigate how coordinated reinforcement learning and explicit reward design influence the trade-off between operational carbon emissions, occupant comfort, and fairness across buildings in CityLearn.

The specific objectives are to:

1. Implement and compare five control approaches: rule-based control, independent SAC, standard MARLISA, carbon-aware MARLISA, and fair carbon-aware MARLISA.
2. Introduce a carbon-aware reward that penalizes grid imports according to the available carbon-intensity signal.
3. Extend the carbon-aware reward with occupied comfort loss and a shared penalty targeting the worst instantaneous comfort loss and the gap between buildings.
4. Evaluate district-level environmental and energy performance together with building-level comfort outcomes.
5. Examine the limitations of the comparison and identify additional experiments needed to distinguish architecture, comfort, and fairness effects.

The study is guided by three research questions:

- **RQ1:** How do the five control approaches differ in operational carbon emissions, electricity consumption, cost, and district demand characteristics?
- **RQ2:** How does the combined carbon, comfort, and fairness reward affect average discomfort, worst-building discomfort, and cross-building disparity?
- **RQ3:** What trade-offs arise between carbon reduction and the provision of acceptable comfort across buildings?

1.4 Significance of the Study

This study connects sustainable energy management with responsible AI by examining who benefits from a controller's decisions. Carbon reduction is evaluated alongside occupant-facing outcomes, allowing apparent environmental improvements to be assessed in relation to the quality of building services.

The study also provides an interpretable approach to reward design. Its carbon, comfort, and fairness components express distinct priorities that can be inspected and evaluated separately. This makes the consequences of the selected objective more transparent and supports discussion of whether a controller distributes discomfort appropriately.

From a methodological perspective, the project emphasizes that architecture and reward function jointly determine controller behavior. Comparing independent SAC with standard MARLISA under different rewards evaluates complete control approaches. Additional

controlled comparisons are required before attributing an observed difference specifically to coordination. Recognizing this distinction improves the soundness of the experimental interpretation.

The intended contribution is an empirical evaluation of a specific comfort and fairness extension to MARLISA. Its value lies in characterizing the resulting trade-offs and identifying the conditions under which the proposed objective is useful.

1.5 Scope of the Study

The project uses CityLearn version 3.0.2 and a scenario containing three buildings. Each building exposes continuous actions for domestic-hot-water storage, electrical storage, and cooling-device power. The principal comparison includes a fixed rule-based reference, independent SAC, and three MARLISA reward configurations.

Fairness is defined at the building level using a temperature-based comfort proxy and differences in building discomfort outcomes. The study does not measure demographic equity or subjective occupant satisfaction. Its carbon objective represents operational emissions associated with simulated grid imports rather than lifecycle emissions.

The existing implementation provides an initial experimental comparison. Longer training, multiple random seeds, held-out evaluation, and reward ablations are needed to support stronger conclusions about generalization and the individual contribution of fairness.

2 Literature Review

2.1 CityLearn as a Building Control Benchmark

Vázquez-Canteli et al. introduced CityLearn to support standardized research in multi-agent reinforcement learning for demand response and urban energy management [1]. A common environment enables researchers to implement controllers using shared interfaces and compare their performance under consistent scenarios. This provides a suitable foundation for the present project's comparison of rule-based, independent, and coordinated control.

CityLearn v2 expanded the environment to include thermal dynamics, occupant interaction, resilience, and carbon-aware objectives [2]. These capabilities make it possible to assess whether flexible building operation preserves comfort while improving energy performance. For a project that controls cooling power, this is particularly relevant because actions can affect both electricity consumption and indoor conditions. The environment therefore supports a broader definition of controller performance than energy savings alone.

The CityLearn v3 preprint further emphasizes service delivery and participant-level outcomes, incorporating features such as electrical constraints, changing assets, and equipment or data failures [5]. Its evaluation perspective reinforces the need to interpret aggregate improvements alongside service shortfalls. However, comparisons must remain specific to the features enabled in the selected scenario; using a recent software version does not imply that every newly available feature is active.

2.2 Reinforcement Learning for Continuous Building Control

Haarnoja et al. developed SAC as an off-policy actor-critic algorithm based on maximum-entropy reinforcement learning [3]. The actor learns a stochastic policy, while the critic estimates the value of actions. Reusing stored transitions supports learning from previous experience, and the entropy term encourages exploration. These properties motivate SAC's use for continuous building-control actions. Nevertheless, the original algorithm's performance does not establish its superiority for every building scenario or training budget.

Kathirgamanathan et al. applied centralized SAC to district demand-side management in CityLearn [6]. Their approach used one agent to control the district, demonstrating an alternative to separate building policies. Centralized and independent control differ in information availability and policy structure, so evidence from centralized SAC should not be treated as direct validation of an independent SAC implementation. In the present study, independent SAC provides a learning-based reference against which coordinated approaches can be assessed.

2.3 Coordination Across Building Agents

MARLISA introduces iterative sequential action selection for coordinated building control [4]. Its CityLearn implementation combines SAC-based building policies with electricity-demand estimation and shared coordination information. Sequential action selection allows updated estimates associated with earlier selections to inform subsequent decisions. This provides each building agent with information about the wider district while retaining separate policies.

Pinto et al. investigated cooperative and coordinated architectures for grid-interactive building energy management [7]. Their work treats the control architecture as an experimental variable and highlights the importance of comparable objectives when assessing architectural differences. It supports examining information sharing and coordination under controlled conditions rather than assuming that a coordinated method will always perform better.

For the present project, coordination and reward coupling are related but distinct. Standard MARLISA uses a district-coupled reward, whereas independent SAC uses a local import penalty. Their comparison consequently measures the behavior of complete approaches. A same-reward comparison, supplemented by a MARLISA information-sharing ablation, would provide stronger evidence about the contribution of coordination.

2.4 Carbon-Aware and Occupant-Centric Reward Design

Reward design determines which outcomes an RL controller is encouraged to optimize. Electricity consumption, carbon emissions, cost, and thermal comfort represent different objectives. A carbon-aware reward weights grid imports by carbon intensity, introducing an incentive to consider when electricity is consumed. When cooling power is controllable, however, reduced imports may also accompany reduced comfort provision. Environmental metrics must therefore be interpreted together with comfort outcomes.

CityLearn v2 demonstrates control tasks involving emissions, demand peaks, and comfort, illustrating how the selected objective influences controller behavior [2]. Recent PIRS work investigates a Predicted Mean Vote-based comfort channel within a SAC reward and uses a matched-weight comparison to examine the effect of comfort representation [8]. The reviewed manuscript provides a useful example of controlled reward evaluation, although its reported results do not establish consistent superiority over rule-based control. Its comfort model also requires inputs or assumptions beyond indoor air temperature.

The present project uses an observable temperature-deviation proxy rather than a physiological comfort model. This keeps the reward connected to the available simulation variables, while limiting the interpretation to setpoint-based comfort. Comparisons should use common external performance indicators because total rewards with different definitions are not directly comparable.

2.5 Fairness in Sequential Decision-Making

Ju, Ghosh, and Shroff investigate fairness in multi-agent Markov decision processes using objectives defined over agents' expected cumulative returns [9]. Their analysis distinguishes aggregate reward maximization from objectives such as max-min and proportional fairness. This provides a conceptual basis for examining whether a controller protects poorly served participants rather than only improving the district total.

The project's shared penalty targets the worst instantaneous comfort loss and the difference between the highest and lowest losses. It draws on the motivation of protecting disadvantaged agents, but differs from fairness defined over cumulative returns. Its instantaneous penalty does not directly guarantee equal episode-level discomfort or acceptable comfort for every building.

OccuReward examines demographic comfort equity in CityLearn through occupant profiles and LLM-assisted reward refinement [10]. Its reported findings indicate that explicit equity feedback can influence the distribution of comfort outcomes. This is relevant recent work, but its demographic focus differs from the present study's building-level objective. It also demonstrates that fairness-aware building reward design is an existing research direction, making the project's contribution specific to its MARLISA extension and evaluation.

Achiam et al.'s constrained-policy formulation offers another perspective: service requirements can be expressed as constraints rather than incorporated only through reward penalties [11]. A weighted reward permits trade-offs and cannot guarantee compliance with a comfort threshold. This motivates evaluating per-building service requirements even when the implemented controller remains an unconstrained SAC-based method.

2.6 Baselines, Generalization, and Reliable Evaluation

The CityLearn Challenge 2022 report describes successful approaches involving forecasting, optimization, reinforcement learning, and hybrid control [12]. Its findings show that effective demand-response control is not limited to a particular RL architecture. The challenge's battery-focused setting also differs from cooling-inclusive control, limiting direct transfer of its performance conclusions.

CHESCA, the winning approach in the 2023 control track, combines building-level optimization with community-level refinement and emphasizes generalization under limited data [13]. It motivates considering stronger classical or hybrid references when assessing learning-based controllers. A fixed BasicRBC schedule represents one reference policy rather than the full range of rule-based control performance.

MERLIN investigates offline and transfer learning to address data requirements and generalization in building communities [14]. Its findings support treating training efficiency and transfer as relevant evaluation concerns, while remaining specific to the studied control tasks. Agarwal et al. further show that few-run point estimates can produce unreliable RL comparisons [15]. Together, these studies motivate longer training, multiple seeds, explicit evaluation splits, and reporting uncertainty alongside average performance.

2.7 Research Gap and Positioning of the Project

The reviewed literature establishes building-control benchmarks, continuous-action RL methods, coordination mechanisms, and fairness-aware objectives. The present project brings these themes together through a specific extension of MARLISA that combines operational carbon penalties with occupied comfort loss and a shared worst-building and disparity penalty.

The study examines whether this combined objective produces a useful carbon–comfort trade-off across buildings. Its evaluation considers environmental performance, service quality, and inequality together, addressing the limitation of relying on district averages or discomfort gaps alone.

The contribution is therefore an application and empirical analysis of an explicit reward formulation within an established coordination framework. The current five-controller comparison evaluates complete approaches. Further comparisons between carbon-only, carbon-plus-comfort, and carbon-plus-comfort-plus-fairness rewards would be required to establish the additional effect of the fairness component.

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